

Fund Flow Implementation

This document describes the fundamental fund flow, lifecycle, and account treatments implemented in the FundFlow application, based on the rules defined in `prompts.txt`.

Account Structure

Account	Type	Purpose
Master Bank	Master	Receives imported bank statement transactions
Master Cash	Master	Mirrors selected bank transactions; reflects total cash at hand
Master Fund	Master	Reflects total fund value (contributions + repayments – disbursements)
Member Cash	Per-member	Individual member's cash balance
Member Fund	Per-member	Individual member's fund balance (can go negative with active loan)

Fund Flow Lifecycle

1. Bank Statement Import

Members transfer money to external bank accounts. The admin regularly imports bank transaction statements (CSV) into the Master Bank account.

- CSV format is configurable via **Settings** → **Bank Import** (delimiter, date format, column mapping, header/skip rows)
- Duplicate detection via transaction hash (date + description + amount + reference)
- Each import creates a **BankStatement** record tracking filename, row counts, and status

Service: `BankImportService::importCsv()`

2. Mirror to Master Cash

The admin selects imported bank transactions and mirrors them to the Master Cash account. Both Master Bank and Master Cash are updated.

- Credits (positive amounts): contributions, deposits, repayments
- Debits (negative amounts): loan disbursements

Service: `FundFlowService::mirrorToCash()`

Status transition: `imported` → `mirrored`

3. Post to Member Cash

The admin posts mirrored transactions to individual member cash accounts. This reflects the credit/debit in the member's cash account as a mirror — no actual debit of Master Cash occurs.

Service: `FundFlowService::postToMember()`

Status transition: `mirrored` → `posted`

4. Contribution Cycle

A contribution cycle starts on a configurable day of the month (default: 6th, via **Settings** → **Contributions**) and ends the day before the next cycle starts.

When the admin runs the contribution cycle collection:

1. **Debit** Member Cash account
2. **Debit** Master Cash (**contribution mirror**)
3. **Credit** Member Fund account
4. **Mirror (credit)** Master Fund account

Only one contribution or repayment per member per cycle is allowed.

Service: `ContributionService::postContribution()`

5. Loan Disbursement

When a loan is disbursed to a member (ledger only — no bank payout yet):

1. **Debit** Loan account (principal)
2. **Debit** Master Fund (**master funded**)
3. **Debit** Member Fund (**member mirror**) — may go negative
4. **Credit** Master Cash (**cash payout mirror**)
5. **Credit** Member Cash (**cash payout**)

Proceeds stay in the member cash account until the member submits a **cash-out request**. Master Cash and Member Cash must move together so **MasterAccountInvariantService** stays balanced.

Service: `LoanService::disburseLoan() → LoanLedgerService::postPartialLoanDisbursement()`

6. Member Cash-Out

When a member requests withdrawal and admin **accepts**:

1. **Debit** Member Cash (**cash out**)
2. **Debit** Master Cash (**cash out mirror**)
3. Create an **uncleared BankTransaction** (negative amount) linked to the request

Clearance (later): Admin matches the uncleared line to an imported bank statement row (**Bank Accounts** → **Statement lines** → **Clear / Match**). No extra ledger entries on match — only **is_cleared** flags update.

Services: `MemberCashOutService::submit() / accept() / clearTransaction()`

7. Loan Repayment

Repayments are processed via the contribution cycle:

1. **Debit** Member Cash
2. **Credit** Member Fund
3. **Mirror (credit)** Master Fund
4. Update loan **total_repaid**; mark completed when fully repaid

Service: `LoanService::recordRepayment()`

Account Balance Rules

- **Master Fund** = sum of all contributions + repayments — loan disbursements (fund leg)
- **Master Cash** = sum of all member cash account balances (keep member/master cash credits and debits paired)
- **Member Fund** may go negative (indicates outstanding loan to be repaid)
- Members may have only one active loan at any given time
- **Bank clearance** reconciles ledger intent to imported bank lines; it does not replace paired master/member postings

Database Tables

bank_statements

Tracks each imported CSV file.

Column	Type	Description
filename	string	Original CSV filename
statement_date	date	Derived from latest transaction date
bank_name	string	Optional bank identifier
total_rows	int	Total data rows in CSV
imported_rows	int	Successfully imported rows
duplicate_rows	int	Skipped duplicate rows
status	enum	pending, processing, completed, failed
imported_by	FK → users	Admin who performed the import
imported_at	timestamp	When the import occurred

bank_transactions

Individual transactions from imported bank statements.

Column	Type	Description
bank_statement_id	FK → bank_statements	Parent statement
transaction_date	date	Transaction date
description	string	Transaction description
amount	decimal(15,2)	Positive = credit, negative = debit
reference	string	Transaction reference/check number
status	enum	imported, mirrored, posted, ignored
member_id	FK → members	Assigned member (set on posting)
hash	string (unique)	Duplicate detection hash
raw_data	text	Original CSV row as JSON

Settings

Group	Key	Default	Description
contribution	cycle_start_day	6	Day of month when cycle starts (1–28)
bank	csv_template	JSON	CSV parsing configuration (delimiter, date format, columns, etc.)

Admin Workflow (Filament UI)

See also [master-bank-manual-credits-and-fund-flow.md](#) for manual master bank ledger entries vs statement import / fund posting workflows.

Banking → Bank Accounts

- **Import statement** (page header, any tab): upload CSV, choose template, optionally specify bank name
- **Statements** tab: view past imports and open statement details
- **Statement lines** tab: work imported rows (mirror, post to member)

Banking → Bank Transactions

- **Mirror to Cash** (row/bulk): move imported transactions to mirrored status, update Master Bank and Master Cash
- **Post to Member** (row): assign a mirrored transaction to a member, update their cash account
- **Ignore** (row/bulk): mark transactions as ignored

Fund Management → Contributions

- **Generate Monthly**: create pending contributions for all active members
- **Post** (row/bulk): run the contribution cycle transfer

Fund Management → Deposits / Cash outs

- **Deposits**: member submits → admin accept credits **master cash + member cash** → uncleared bank line until matched
- **Cash outs**: member submits → admin accept debits **master cash + member cash** → uncleared bank line until matched

Loans

- **Approve → Disburse**: fund debits + paired cash credits (member + master)
- **Cash out** (member portal): separate step when the member wants funds sent to their bank
- **Mark bank payout** (`payoutLoan()`): optional timestamp only; ledger payout is via cash-out + bank clearance